# BANGALORE INSTITUTE OF TECHNOLOGY

An autonomous institute affiliated to **VTU**

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*"Technologyfor Prosperity"*

*[ESTD:1979]*

## ಬೆಂಗಳೂರು ತಾಂತ್ರಿಕ ಮಹಾವಿದ್ಯಾಲಯ

*(ಸ್ವಾಯತ್ತ ಸಂಸ್ಥೆ)*

*"ಸಮೃದ್ಧಿಗಾಗಿ ತಂತ್ರಜ್ಞಾನ"*

**ಬೆಂಗಳೂರು- *560004***

B I T



# CSE

## COMPUTER SCIENCE & ENGINEERING

### SCHEME & SYLLABUS

### V SEMESTER

### 2024 BATCH

**Scheme of Teaching and Examinations-2025**
**Outcome-Based Education (OBE) and Choice-Based Credit System (CBCS)**
**(Effective from the academic year 2025-26)**

**ASSESSMENT DETAILS FOR CONTINUOUS INTERNAL EVALUATION (CIE) AND SEMESTER END EXAM (SEE)**

| 1 | INTEGRATED COURSE (IC) WITH 3 or 4 CREDITS |
|---|---|

- Integrated Course (IC) combines theory with practical components.
- Theory components shall be evaluated through both Continuous Internal Evaluation (CIE) and Semester End Examination (SEE).
- The practical component shall be assessed only through CIE.
- The IC course is assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 50% of the maximum marks, i.e., 50 marks out of 100 and for Semester End Exam (SEE) is also 50% of the maximum marks, i.e., 50 marks out of 100.
- To qualify and become eligible to appear for SEE, the minimum passing mark for the CIE is 40% of the maximum marks, i.e., 20 marks out of 50.
- The minimum passing mark for the SEE is 35% of the maximum marks, i.e., 18 marks out of 50.
- A student is deemed to have successfully completed the course if the combined total of CIE and SEE is at least 40 out of 100 marks.

**CONTINUOUS INTERNAL EVALUATION (CIE)**

- The CIE marks for the theory component shall be 25 marks, and for the practical component, 25 marks.
- The minimum marks to be secured in the CIE to appear for SEE shall be 40% of maximum marks, i.e., 10 marks out of 25 marks in the theory component and 40% of maximum marks i.e. 10 marks out of 25 marks in the practical component.

**CIE for the theory component of the IC**

- Two Unit Tests, each of 50 Marks and duration of 1.5 hour, shall be conducted. The first test will be conducted after 50% of the syllabus is completed and the second test after 100% of the syllabus is completed.
- Total marks scored from the two unit tests out of 100 shall be scaled down to 25 marks.

- If necessary, one improvement unit test may be conducted after the second unit test. However, the sum of the best two, unit test marks shall be scaled down to 25 marks.

**CIE for the practical component of the IC**

- The CIE marks for the practical component shall be 25, with 15 marks allocated for continuous internal evaluation of each experiment and 10 marks for a test conducted at the end of the semester.
- Each laboratory experiment is assessed using appropriate rubrics. The summation of all the experiments marks to be scaled down to 15 marks.
- A laboratory test shall be conducted either at the end of the final week of the semester or upon completion of all prescribed experiments, whichever occurs earlier. The test shall be evaluated for 50 marks, which will subsequently be scaled down to 10 marks. In the CIE test, each student shall be required to perform one experiment from Part A and one experiment from Part B.
- The minimum marks to be secured shall be 40% of the maximum marks i.e., 10 out of 25.

**SEMESTER END EXAMINATION (SEE)**

- The Theory SEE is conducted according to the scheduled timetable with a common question paper for each course.
- The question paper shall be set for 100 marks, with English as the medium of the paper. The duration of the SEE is 3 hours.
- The question paper consists of 10 questions with two main questions from each module. Each main question is set for 20 marks with a maximum of 3 sub-questions with mix of topics within the module. The students must answer 5 full questions, selecting one full question from each module.
- The marks scored by the student out of 100 are scaled down to 50 marks.

| Integrated Course (IC) with 3/4 Credits | | | | | | |
|---|---|---|---|---|---|---|
| **Evaluation Type** | | | **Assessment** | **Test/Exam Marks Allocation** | **Scaled down Marks** | **Minimum Marks to be Scored** |
| **CONTINUOUS INTERNAL EVALUATION (CIE) ( 50 Marks)** | Theory Component | Internal Assessment (IA) Tests | Test 1 (1.5 hour) | 50 | 25 | 10 |
| | | | Test 2 (1.5 hour) | 50 | | |
| | | **Total CIE Theory** | | | **25** | **10** |
| | Practical Component | Regular Practical Class | | As per rubrics | 15 | 10 |
| | | Practical Test | | 50 | 10 | |
| | | **Total CIE Practical** | | | **25** | **10** |
| **Total CIE (Theory + Practical component)** | | | | | **50** | **20** |
| **SEMESTER END EXAMINATION (SEE) (50 Marks)** | | | Theory examination (3 hours) | 100 | 50 | 18 |
| **TOTAL (CIE + SEE) MARKS** | | | | | **100** | **40** |

## 2 APPLIED SCIENCE COURSE (ASC)/ PROGRAMME SPECIFIC COURSE (PCC) / ENGINEERING SCIENCE COURSE (ESC) WITH 04 OR 03 CREDITS.

- These courses are assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 50% of the maximum marks, i.e., 50 marks out of 100 and for Semester End Exam (SEE) is also 50% of the maximum marks, i.e., 50 marks out of 100.
- To qualify and become eligible to appear for SEE, the minimum passing mark for the CIE is 40% of the maximum marks, i.e., 20 marks out of 50.
- The minimum passing mark for the SEE is 35% of the maximum marks, i.e., 18 marks out of 50.
- A student is deemed to have successfully completed the course if the combined total of CIE and SEE is at least 40 out of 100 marks.

### CONTINUOUS INTERNAL EVALUATION (CIE)

- The CIE marks for the unit tests shall be 25 marks, and for the Continuous Comprehensive Assessments (CCA), 25 marks.
- Two Unit Tests, each of 50 Marks and duration of 1.5 hour, shall be conducted. The first test will be conducted after 50% of the syllabus is completed and the second test after 100% of the syllabus is completed.

- Total marks scored from the two unit tests out of 100 shall be scaled down to 25 marks.
- If necessary, one improvement unit test may be conducted after the second unit test. However, the sum of the best two, unit test marks shall be scaled down to 25 marks.
- Two Continuous Comprehensive Assessments (CCA), each for 25 marks, are to be conducted.
- It is recommended to include any two learning activities aimed at enhancing the holistic development of students. These activities should align with course objectives and promote higher-order thinking and application-based learning.
- Total marks scored from two CCAs, out of 50 marks, shall be scaled down to 25 marks.

**SEMESTER END EXAMINATION (SEE)**

- The Theory SEE is conducted according to the scheduled timetable with a common question paper for each course.
- The question paper shall be set for 100 marks, with English as the medium of the paper. The duration of the SEE is 3 hours.
- The question paper consists of 10 questions with two main questions from each module. Each main question is set for 20 marks with a maximum of 3 sub-questions with mix of topics within the module. The students must answer 5 full questions, selecting one full question from each module.
- The marks scored by the student out of 100 are scaled down to 50 marks.

<table>
<tr><th colspan="6">APPLIED SCIENCE COURSE (ASC)/ PROGRAMME SPECIFIC COURSE (PCC) / ENGINEERING SCIENCE COURSE (ESC) WITH 04 OR 03 CREDITS.</th></tr>
<tr><th colspan="3">Evaluation Type</th><th>Assessment</th><th>Test/Exam Marks Allocation</th><th>Scaled down Marks</th><th>Minimum Marks to be Scored</th></tr>
<tr><td rowspan="4">CONTINUOUS INTERNAL EVALUATION (CIE)<br>( 50 Marks)</td><td rowspan="2">Internal Assessment (IA) Tests</td><td>Test 1 (1.5 hour)</td><td>50</td><td rowspan="2">25</td><td rowspan="2">10</td></tr>
<tr><td>Test 2 (1.5 hour)</td><td>50</td></tr>
<tr><td rowspan="2">Continuous Comprehensive Assessment (CCA)</td><td>CCA-1</td><td>25</td><td rowspan="2">25</td><td rowspan="2">10</td></tr>
<tr><td>CCA-2</td><td>25</td></tr>
<tr><td colspan="4">Total CIE Marks</td><td></td><td>50</td><td>20</td></tr>
<tr><td colspan="2">SEMESTER END EXAMINATION (SEE)<br>(50 Marks)</td><td>Theory Examination (3 hours)</td><td>100</td><td>50</td><td>18</td></tr>
<tr><td colspan="4">Total SEE Marks</td><td></td><td>50</td><td>18</td></tr>
<tr><td colspan="4">TOTAL (CIE + SEE) MARKS</td><td></td><td>100</td><td>40</td></tr>
</table>

| 3 | COMPUTER AIDED ENGINEERING DRAWING (STREAM SPECIFIC) WITH 3 CREDITS. |
|---|---|

- Computer Aided Engineering Drawing course is assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 50% of the maximum marks, i.e., 50 marks out of 100 and for Semester End Exam (SEE) is also 50% of the maximum marks, i.e., 50 marks out of 100.
- To qualify and become eligible to appear for SEE, the minimum passing mark for the CIE is 40% of the maximum marks, i.e., 20 marks out of 50.
- The minimum passing mark for the SEE is 35% of the maximum marks, i.e., 18 marks out of 50.
- A student is deemed to have successfully completed the course if the combined total of CIE and SEE is at least 40 out of 100 marks.

**CONTINUOUS INTERNAL EVALUATION (CIE)**

- Continuous Internal Evaluation shall be for 50 marks, with 25 marks allocated for continuous internal evaluation of class work and 25 marks for a test conducted at the end of the semester.
- The final CIE (50) = Class work marks (25) + Test marks (25)
- Class work marks should comprise of continuous evaluation of Drawing work of students with a model-wise distribution as shown in the table below for 100 marks. Marks obtained out of 100 shall be scaled down to 25 marks.
- Towards the end of the semester, a laboratory test is conducted for 100 marks, following the same pattern as the of Semester End examination (SEE) shown in the table below. Marks obtained out of 100 shall be scaled-down to 25 marks.

**SEMESTER END EXAMINATION (SEE)**

- The Practical Semester End Examination (SEE) is conducted batch-wise according to the scheduled timetable.
- The question paper shall be set for 100 marks, with English as the medium of the paper. The duration of the SEE is 3 hours.

- Two full questions shall be set from Modules 1, 2, 3 and 4. Students need to answer one full question from each module.
- Two full questions set from each Module shall cover the entire topic of the respective module.
- SEE shall be conducted by one Internal and one External Examiner.
- Evaluation shall be carried out jointly by both the examiners.
- The student may be awarded full marks, if he/she completes a solution on computer display without sketch.
- The weightage and distribution of marks for each Module is as shown in the following table.
- The marks scored by the student out of 100 shall be scaled down to 50 marks.

| **Computer Aided Engineering Drawing (Stream Specific) with 3 Credits.** | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Evaluation Type** | | **Module** | **Marks allotted** | **Computer Display & Printout (a)** | Sketching (b) | **Total (a+b)** | **Scaled down Marks** | **Minimum Marks to be Scored** |
| **CONTINUOUS INTERNAL EVALUATION (CIE) ( 50 Marks)** | Class work | Module 1 | 20 | 15 | 5 | 100 | 25 | 10 |
| | | Module 2 | 20 | 15 | 5 | | | |
| | | Module 3 | 20 | 15 | 5 | | | |
| | | Module 4 | 20 | 15 | 5 | | | |
| | | Module 5 | 20 | 15 | 5 | | | |
| | Laboratory Test (Pattern to be same as SEE) | | | | | 100 | 25 | 10 |
| **Total CIE Marks** | | | | | | | **50** | **20** |
| **SEMESTER END EXAMINATION (SEE) (50 Marks)** (3 hours) | | Module 1 | 20 | 15 | 5 | **100** | **50** | **18** |
| | | Module 2 | 30 | 25 | 5 | | | |
| | | Module 3 | 25 | 20 | 5 | | | |
| | | Module 4 | 25 | 20 | 5 | | | |
| **Total SEE Marks** | | | | | | **100** | **50** | **18** |
| **TOTAL (CIE + SEE) MARKS** | | | | | | | **100** | **40** |

## 4 PROGRAM-SPECIFIC COURSE LAB WITH 1 CREDIT

- The Program-Specific Course Lab is assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 50% of the maximum marks, i.e., 50 marks out of 100 and for Semester End Exam (SEE) is also 50% of the maximum marks, i.e., 50 marks out of 100.
- To qualify and become eligible to appear for SEE, the minimum passing mark for the CIE is 40% of the maximum marks, i.e., 20 marks out of 50.
- The minimum passing mark for the SEE is 35% of the maximum marks, i.e., 18 marks out of 50.

- A student is deemed to have successfully completed the course if the combined total of CIE and SEE is at least 40 out of 100 marks.

**CONTINUOUS INTERNAL EVALUATION (CIE)**

- The CIE marks awarded shall be based on the continuous evaluation of the laboratory report using a defined set of rubrics. Each experiment report can be evaluated for 25 marks. The summation of all the experiments marks to be scaled down to 25 marks.
- A laboratory test of three hours' duration shall be conducted either at the end of the final week of the semester or upon completion of all prescribed experiments, whichever occurs earlier. The test shall follow the pattern of the Semester End Examination (SEE) and shall be evaluated for 50 marks, which will subsequently be scaled down to 25 marks. For both the CIE test and the SEE, each student shall be required to perform one experiment from Part A and one experiment from Part B.

**SEMESTER END EXAMINATION (SEE)**

- The Practical Semester End Examination (SEE) is conducted batch-wise according to the scheduled timetable.
- The question paper shall be set for 100 marks, with English as the medium of the paper. The duration of the SEE is 3 hours.
- SEE shall be conducted by one Internal and one External Examiner.
- Evaluation shall be carried out jointly by both the examiners.
- The marks scored by the student out of 100 shall be scaled down to 50 marks.

| **Program-Specific Course Lab with 1 credit** | | | | | |
|---|---|---|---|---|---|
| **Evaluation Type** | | **Assessment** | **Test/Exam Marks Allocation** | **Scaled down Marks** | **Minimum Marks to be Scored** |
| **CONTINUOUS INTERNAL EVALUATION (CIE) ( 50 Marks)** | Regular Laboratory Class | As per rubrics | 30 for each experiment | 25 | 20 |
| | Laboratory Test | Pattern to be same as SEE | 50 | 25 | |
| | **Total CIE Practical** | | | **50** | **20** |
| **SEMESTER END EXAMINATION (SEE) (50 Marks)** | | Practical examination (3 hours) | 100 | 50 | 18 |
| **TOTAL (CIE + SEE) MARKS** | | | | **100** | **40** |

## 5 INNOVATION & DESIGN THINKING LAB WITH 1 CREDIT

- The Innovation & Design Thinking Lab is assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 50% of the maximum marks, i.e., 50 marks out of 100 and for Semester End Exam (SEE) is also 50% of the maximum marks, i.e., 50 marks out of 100.
- To qualify and become eligible to appear for SEE, the minimum passing mark for the CIE is 40% of the maximum marks, i.e., 20 marks out of 50.
- The minimum passing mark for the SEE is 35% of the maximum marks, i.e., 18 marks out of 50.
- A student is deemed to have successfully completed the course if the combined total of CIE and SEE is at least 40 out of 100 marks.

**CONTINUOUS INTERNAL EVALUATION (CIE)**

- The CIE marks awarded shall be based on the continuous evaluation of the allocation Parameters for Social Entrepreneurship, Innovation & Design Thinking using Atal Idea/Tinkering Lab or Maker Space. The weightage and distribution of marks for each component is as shown in the following table.
- The summation of all the components marks shall be the CIE marks for 50.

| CIE Parameters (50 Marks) | | | |
|---|---|---|---|
| **Sl. No.** | **CIE Component/Week** | **Marks** | **Description** |
| 1 | Orientation Activities & Communication Skills | 5 | Participation in Week 1–3 orientation, communication and teamwork skill-building exercises. |
| 2 | Empathy & Field Exploration Documentation | 10 | Quality and completeness of field visit reflections, stakeholder interviews, and activity book. |
| 3 | Problem Definition and Framing | 10 | Clarity of challenge statements, use of "How Might We", Affinity Mapping, Problem Trees. |
| 4 | Ideation & Mind Mapping | 5 | Participation in brainstorming, mind mapping, idea filtering sessions. |
| 5 | Prototype Development & Iteration | 10 | Quality and creativity of prototype/model, user testing, feedback collection, iterations. |
| 6 | Final Presentation & Pitch | 5 | Project pitching, poster presentation, storytelling and scalability model. |
| 7 | Teamwork, Journal, and Engagement | 5 | Peer and mentor evaluation of participation, teamwork, journal updates. |
| 8 | Total CIE marks | 50 | Final CIE marks to be considered |

**SEMESTER END EXAMINATION (SEE)**

- The Practical Semester End Examination (SEE) is conducted batch-wise according to the scheduled timetable.
- The examination shall be for 100 marks. The duration of the SEE is 3 hours.
- The students will exhibit their projects along with the presentation and Viva -voce. The weightage and distribution of marks for each component is as shown in the following table.
- SEE shall be conducted by one Internal and one External Examiner.
- Evaluation shall be carried out jointly by both the examiners.
- The marks scored by the student out of 100 shall be scaled down to 50 marks.

| SEE Parameters (50 Marks) | | | |
|---|---|---|---|
| **Sl. No.** | **Evaluation Parameter** | **Marks** | **Details** |
| 1 | Prototype / Solution Demonstration | 30 | Working functionality, creativity, use of lab tools, relevance to the problem. |
| 2 | Final Presentation / Social Pitch | 20 | Clarity, storytelling, problem-solution fit, communication, visual aids. |
| 3 | Business Model or Sustainability Plan | 10 | Feasibility, cost-effectiveness, scalability, and alignment with SDGs. |
| 4 | Viva Voce | 20 | Individual understanding, contribution, tools used, learning outcomes. |
| 5 | Documentation Report / Portfolio | 20 | Project report, reflection, team activity log, stakeholder input summaries. |
| 6 | Total SEE marks | 50 | |

## 6 HUMANITY AND SOCIAL SCIENCE AND MANAGEMENT COURSE (HSMS) WITH 01 CREDIT.

- The HSMS course is assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 50% of the maximum marks, i.e., 50 marks out of 100 and for Semester End Exam (SEE) is also 50% of the maximum marks, i.e., 50 marks out of 100.
- The minimum passing mark for the CIE is 40% of the maximum marks, i.e., 20 marks out of 50.

- The minimum passing mark for the SEE is 35% of the maximum marks, i.e., 18 marks out of 50.
- A student is deemed to have satisfied the academic requirements and earned the credits allotted to each course if the student secures not less than 35%, i.e., 18 Marks out of 50 in the SEE, and a minimum of 40%, i.e., 40 marks out of 100 in the total of the CIE and SEE combined.

**CONTINUOUS INTERNAL EVALUATION (CIE)**

- Two Unit Tests, each of 25 Marks and duration of 1 hour, shall be conducted. The first test will be conducted after 50% of the syllabus is completed and the second test after 100% of the syllabus is completed. The question paper for this course shall consist of Multiple Choice questions (MCQs) with 1 mark each.
- Total marks scored from the two unit tests out of 50 shall be scaled down to 25 marks.
- If necessary, one improvement unit test may be conducted after the second unit test. However, the sum of the best two unit test marks shall be scaled down to 25 marks.

**Comprehensive Continuous Assessment (CCA) Methods**

- Two Continuous Comprehensive Assessments (CCA), each for 25 marks, are to be conducted.
- It is recommended to include any two learning activities aimed at enhancing the holistic development of students. These activities should align with course objectives and promote higher-order thinking and application-based learning.
- Total marks scored from two CCAs, out of 50 marks, shall be scaled down to 25 marks.

**SEMESTER END EXAMINATION (SEE)**

- The Theory SEE is conducted according to the scheduled timetable with a common question paper for each course.
- The question paper shall be set for 50 marks. The medium of the question paper for Samskrutika Kannada - 1BKSK109/ 1BKSK209 and Balake Kannada - 1BKBK109/ 1BKBK209 shall be Kannada.
- The question paper for this course shall consist of 50 Multiple Choice questions (MCQs) with 1 mark each with equal weightage to all modules. The students must answer all questions.
- The marks scored by the student out of 50 are considered.

| Humanity and Social Science and Management Course (HSMS) with 01 Credit. | | | | | |
|---|---|---|---|---|---|
| **Evaluation Type** | | **Assessment** | **Test/Exam Marks Allocation** | **Scaled down Marks** | **Minimum Marks to be Scored** |
| **CONTINUOUS INTERNAL EVALUATION (CIE) ( 50 Marks)** | Internal Assessment (IA) Tests (Multiple Choice questions (MCQs)) | Test 1 (1 hour) | 25 | 25 | 20 |
| | | Test 2 (1 hour) | 25 | | |
| | Continuous Comprehensive Assessment (CCA) | CCA-1 | 25 | 25 | |
| | | CCA-2 | 25 | | |
| **Total CIE Marks** | | | | **50** | **20** |
| **SEMESTER END EXAMINATION (SEE) (50 Marks)** (Multiple Choice questions (MCQs)) | | Theory Examination (1 hour) | 50 | 50 | 18 |
| **Total SEE Marks** | | | | **50** | **18** |
| **TOTAL (CIE + SEE) MARKS** | | | | **100** | **40** |

## 7 COMMUNICATION SKILLS (1BENG106/1BENG206); ABILITY ENHANCEMENT COURSE (AEC) WITH 01 CREDIT.

- The COMMUNICATION SKILLS (1BENG106/1BENG206) course is assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 50% of the maximum marks, i.e., 50 marks out of 100 and for Semester End Exam (SEE) is also 50% of the maximum marks, i.e., 50 marks out of 100.
- The minimum passing mark for the CIE is 40% of the maximum marks, i.e., 20 marks out of 50.
- The minimum passing mark for the SEE is 35% of the maximum marks, i.e., 18 marks out of 50.
- A student is deemed to have satisfied the academic requirements and earned the credits allotted to each course if the student secures not less than 35%, i.e., 18 Marks out of 50 in the SEE, and a minimum of 40%, i.e., 40 marks out of 100 in the total of the CIE and SEE combined.

**CONTINUOUS INTERNAL EVALUATION (CIE)**

- Two Unit Tests, each of 25 Marks and duration of 1 hour, shall be conducted. The first test will be conducted after 50% of the syllabus is completed and the second test after 100% of the syllabus is completed. The question paper for this course shall be descriptive type with the pattern similar to SEE.
- Total marks scored from the two unit tests out of 50 shall be scaled down to 25 marks.

- If necessary, one improvement unit test may be conducted after the second unit test. However, the sum of the best two unit test marks shall be scaled down to 25 marks.

**Comprehensive Continuous Assessment (CCA) Methods**

- Two Continuous Comprehensive Assessments (CCA), each for 25 marks, are to be conducted.
- It is recommended to include any two learning activities aimed at enhancing the holistic development of students. These activities should align with course objectives and promote higher-order thinking and application-based learning.
- Total marks scored from two CCAs, out of 50 marks, shall be scaled down to 25 marks.

**SEMESTER END EXAMINATION (SEE)**

- The Theory SEE is conducted according to the scheduled timetable with a common question paper for each course.
- The SEE shall be conducted in English, with a duration of two hours and a maximum of 50 marks.
- The question paper shall be descriptive in nature and shall ensure equal weightage to all modules.
- The pattern of the question paper shall be as follows:

  Part A: Compulsory multiple-choice questions carrying 10 marks in total (1 mark each).

  Part B: Compulsory case study questions carrying 10 marks in total.

  Part C: Five questions to be answered, one from each unit, carrying 6 marks each.
- The marks scored by the student out of 50 are considered.

<table>
<tr><th colspan="6">Communication Skills (1BENG106/1BENG206); Ability Enhancement Course (AEC) with 01 Credit.</th></tr>
<tr><th colspan="3">Evaluation Type</th><th>Assessment</th><th>Test/Exam Marks Allocation</th><th>Scaled down Marks</th><th>Minimum Marks to be Scored</th></tr>
<tr><td rowspan="4">CONTINUOUS INTERNAL EVALUATION (CIE) ( 50 Marks)</td><td rowspan="2">Internal Assessment (IA) Tests</td><td>Test 1 (1 hour)</td><td>25</td><td rowspan="2">25</td><td rowspan="4">20</td></tr>
<tr><td>Test 2 (1 hour)</td><td>25</td></tr>
<tr><td rowspan="2">Continuous Comprehensive Assessment (CCA)</td><td>CCA-1</td><td>25</td><td rowspan="2">25</td></tr>
<tr><td>CCA-2</td><td>25</td></tr>
<tr><td colspan="4">Total CIE Marks</td><td>50</td><td>20</td></tr>
<tr><td colspan="2">SEMESTER END EXAMINATION (SEE) (50 Marks)</td><td>Theory Examination (2 hours)</td><td>50</td><td>50</td><td>18</td></tr>
<tr><td colspan="4">Total SEE Marks</td><td>50</td><td>18</td></tr>
<tr><td colspan="4">TOTAL (CIE + SEE) MARKS</td><td>100</td><td>40</td></tr>
</table>

| 8 | ABILITY ENHANCEMENT COURSE (AEC) AS NON-CREDIT MANDATORY COURSE (NCMC). |
|---|---|

- Indian Constitution & Engineering Ethics (1BICO107/ 1BICO207) and Soft Skills (1BSKS106/ 1BSKS206) are Ability Enhancement Courses (AEC) as Non-Credit Mandatory Courses (NCMC)
- These courses are assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 100% of the maximum marks, i.e., 100 marks out of 100.
- The minimum passing mark for the CIE is 40% of the maximum marks, i.e., 40 marks out of 100.
- PP: (Pass/Pass) is assigned to a non-credit course. “PP” represents pass in course provided students have successfully completed the CIE requirements.
- Otherwise, “NP-not pass shall be awarded. “PP” is essential for the award of the degree.
- Two Continuous Comprehensive Assessments (CCA), each for 50 marks, are to be conducted.
- It is recommended to include any two learning activities aimed at enhancing the holistic development of students. These activities should align with course objectives and promote higher-order thinking and application-based learning.
- Total marks scored from two CCAs, out of 100 marks, shall be considered.

| 9 | PROFESSIONAL CORE COURSE LAB (PCCL) WITH 1 CREDIT |
|---|---|

- The Professional Core Course Lab is assessed out of a total of 100 marks.
- The weightage of Continuous Internal Evaluation (CIE) is 50% of the maximum marks, i.e., 50 marks out of 100 and for Semester End Exam (SEE) is also 50% of the maximum marks, i.e., 50 marks out of 100.
- To qualify and become eligible to appear for SEE, the minimum passing mark for the CIE is 40% of the maximum marks, i.e., 20 marks out of 50.
- The minimum passing mark for the SEE is 35% of the maximum marks, i.e., 18 marks out of 50.
- A student is deemed to have successfully completed the course if the combined total of CIE and SEE is at least 40 out of 100 marks.

**CONTINUOUS INTERNAL EVALUATION (CIE)**

- The CIE marks awarded shall be based on the continuous evaluation of the laboratory report using a defined set of rubrics. Each experiment report shall be evaluated for 25 marks. The summation of all the experiments marks to be scaled down to 25 marks.
- A laboratory test of three hours' duration shall be conducted either at the end of the final week of the semester or upon completion of all prescribed experiments, whichever occurs earlier. The test shall follow the pattern of the Semester End Examination (SEE) and shall be evaluated for 50 marks, which will subsequently be scaled down to 25 marks. For both the CIE test and the SEE, each student shall be required to perform one experiment from Part A and one experiment from Part B.

**SEMESTER END EXAMINATION (SEE)**

- The Practical Semester End Examination (SEE) is conducted batch-wise according to the scheduled timetable.
- The question paper shall be set for 100 marks, with English as the medium of the paper. The duration of the SEE is 3 hours.
- SEE shall be conducted by one Internal and one External Examiner.
- Evaluation shall be carried out jointly by both the examiners.
- The marks scored by the student out of 100 shall be scaled down to 50 marks.

| **Professional Core Course Lab with 1 credit** | | | | | |
|---|---|---|---|---|---|
| **Evaluation Type** | | **Assessment** | **Test/Exam Marks Allocation** | **Scaled down Marks** | **Minimum Marks to be Scored** |
| **CONTINUOUS INTERNAL EVALUATION (CIE) ( 50 Marks)** | Regular Laboratory Class | As per rubrics | 30 for each experiment | 25 | 20 |
| | Laboratory Test | Pattern to be same as SEE | 50 | 25 | |
| | **Total CIE Practical** | | | **50** | **20** |
| **SEMESTER END EXAMINATION (SEE) (50 Marks)** | | Practical examination (3 hours) | 100 | 50 | 18 |
| **TOTAL (CIE + SEE) MARKS** | | | | **100** | **40** |

# BANGALORE INSTITUTE OF TECHNOLOGY
## K.R Road, V.V Puram, Bangalore - 560004

**Ref:/BIT/IQAC/015a/2025-26 Dated xx/11/2025**

## Continuous Internal Evaluation Pattern with effect from 2024-25 Batch onwards

| Course Type | Credits | Theory | | | | | | | | | Practical | | | | CIE | Min | SEE | | Final CIE+SEE | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A | | | | B | | | | C | D | E | F | | | | Max | Min | Max | Min |
| | | Test 1 | Test 2 | Scaled down to | Min Marks | CCA 1 | CCA 2 | Scaled down to | Min Marks | Final Marks of A+B | Continous Evaluation | Lab Test | Final marks of D+E | Minimum Marks | Final Marks C+F | CIE Min Marks | | | | |
| IPCC/ASC-IC/PLC-IC/ESC-IC | 3/4 | 50 | 50 | 25 | 10 | | | | | 25 | 15 | 10 | 25 | 10 | 50 | 20 | 50 | 18 | 100 | 40 |
| PCC/BSC/ASC/ETC/ESC/PSC/PLC/PEC/OEC/AEC | 2 /3 /4 | 50 | 50 | 25 | 10 | 25 | 25 | 25 | 10 | 50 | | | | | 50 | 20 | 50 | 18 | 100 | 40 |
| PCCL/PSCL/AEC-L | 1 | | | | | | | | | | 25 | 25 | 50 | 20 | 50 | 20 | 50 | 18 | 100 | 40 |
| ESC-CAED | 3 | | | | | | | | | | 25 | 25 | 50 | 20 | 50 | 20 | 50 | 18 | 100 | 40 |
| AEC/SEC/SDC/UHV (T) | 1 | 25 | 25 | 25 | 10 | 25 | 25 | 25 | 10 | 50 | | | | | 50 | 20 | 50 | 18 | 100 | 40 |
| AEC/SEC/SDC (P) | 1 | | | | | | | | | | 50 | | 50 | 20 | 50 | 20 | 50 | 18 | 100 | 40 |
| UHV (P) | 1 | | | | | | | | | | 100 | | 100 | 40 | 100 | 40 | NA | | | |
| HSMC/HSMS | 1 | 25 | 25 | 25 | 10 | 25 | 25 | 25 | 10 | 50 | | | | | 50 | 20 | 50 | 18 | 100 | 40 |
| NCMC/MC/IKS | 0 | | | | | 50 | 50 | 100 | 40 | 100 | | | | | 100 | | NA | | | |
| MINI/MAJOR-P1 PROJECT | 2 | | | | | | | | | | 100 | | 100 | 50 | 100 | 50 | NA | | | |
| MAJOR-P2 PROJECT | 6 | | | | | | | | | | 100 | | 100 | 50 | 100 | 50 | 100 | 50 | 200 | 100 |
| INTERNSHIP | 10 | | | | | | | | | | 100 | | 100 | 50 | 100 | 50 | 100 | 50 | 200 | 100 |

Note:

CCA Comprehensive Continous Assesment

Ref: VTU/BGM/Aca/BoS/2023/1335 dated 13 June 2023, Any amendments issued by VTU from Time to Time to be followed

| SCHEME | Department of: Computer Science and Engineering |
|---|---|

## V Semester

| Sl. No. | Course Type | Course Code | Course Title | TD/PSB | TLP | | | | Marks | | | Credits |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | L | T | P | Contact Hours | CIE | SEE | Total | |
| 1 | HSMS | BCS501 | Software Engineering & Project Management | TD: CS PSB: CSB | 4 | 0 | 0 | 4 | 50 | 50 | 100 | 4 |
| 2 | IPCC | BCS502 | Computer Networks & Security | TD: CS PSB: CSB | 3 | 0 | 2 | 5 | 50 | 50 | 100 | 4 |
| 3 | PCC | BCS503 | Theory of Computation | TD: CS PSB: CSB | 4 | 0 | 0 | 4 | 50 | 50 | 100 | 4 |
| 4 | PCCL | BCSL504 | Web Technology Lab | TD: CS PSB: CSB | 0 | 0 | 2 | 2 | 50 | 50 | 100 | 1 |
| 5 | PEC | BCS515A | Computer Graphics | TD: CS PSB: CSB | 3 | 0 | 0 | 3 | 50 | 50 | 100 | 3 |
| | | BCS515B | Artificial Intelligence | TD: CS PSB: CSB | | | | | | | | |
| | | BCS515C | Unix System Programming | TD: CS PSB: CSB | | | | | | | | |
| | | BCS515D | Natural Language Processing | TD: CS PSB: CSB | | | | | | | | |
| 6 | PROJ | BCS586 | Minor Project | TD: CS PSB: CS | 0 | 0 | 4 | 4 | 100 | --- | 100 | 2 |
| 7 | AEC | BRMK507 | Research Methodology and IPR | TD: CS PSB: Any | 3 | 0 | 0 | 3 | 50 | 50 | 100 | 3 |
| 8 | MC | BESK508 | Environmental Studies and Waste Management | TD:CS PSB: Any | 1 | 0 | 0 | 1 | 50 | 50 | 100 | 1 |
| 9 | NCMC | BNSK509 | National Service Scheme | NSS | 0 | 0 | 2 | 2 | 100 | ---- | 100 | 0 |
| | | BPEK509 | Physical Education | PE | | | | | | | | |
| | | BYOK509 | Yoga | Yoga | | | | | | | | |
| | | | | Total | | | | 28 | 450 | 350 | 800 | 22 |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | SOFTWARE ENGINEERING & PROJECT MANAGEMENT | BCS501 | 4 | 50 | 4:0:0:0 | 3 | 50 | 50 | PCC (T) | Theory |

## Course Objectives

1. Outline software Process and software engineering principles involved in building large software programs, also understanding the Agile model and other process models for process assessment and improvement.
2. Describe the process of requirement gathering, requirement classification, requirement specification and requirements validation.
3. Comprehend the concepts of software design and strategic approach to Software testing, verification and validation.
4. Recognize the importance of Project Management with its methods and methodologies.
5. Identify Project evaluation and program management measures. Understand the metrics involved in software project estimations.

## Teaching–Learning Process

1. Lecturer method (L) need not be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video/Animation to explain functioning of various concepts.
3. Encourage collaborative (Group Learning) Learning in the class.
4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
5. Adopt Problem Based-Learning (PBL), which fosters student's Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.

## Modern Tools

Jira ,Figma , GitHub, Google Cloud Platform

### Module 1

**Software and Software Engineering**: The nature of Software, The unique nature of WebApps, Software Engineering, The software Process, Software Engineering Practice, Software Myths.
**Process Models**: A generic process model, Process assessment and improvement, Prescriptive process models: Waterfall model, Incremental process models, Evolutionary process models, Concurrent models, Specialized process models. Unified Process, Personal and Team process models.
**Agile Development**: What is Agility? Agility and the cost of change. What is an agile Process? Extreme Programming (XP), Other Agile Process Models, A tool set for Agile process.

**TB1:1.1-1.6,2.1-2.6,3.1-3.6** **RBT: 2** **WK: 1,4**

### Module 2

**Understanding Requirements**: Requirements Engineering, Establishing the ground work, Eliciting Requirements, Developing use cases, Building the requirements model, Negotiating Requirements, Validating Requirements. Requirements Modeling Scenarios Information, Analysis classes.
**Behavior and Patterns**: Requirement Analysis, Scenario based modeling, UML models that supplement the Use Case, Data modeling Concepts, Class Based Modeling.
Requirement Modeling Strategies, Flow oriented Modeling, Behavioral Modeling and Patterns for Requirement Modelling.

**TB1:5.1-5.7,6.1-6.5,7.1-7.4** **RBT: 6** **WK: 3,4,5,6**

### Module 3

**Software Design Concepts**: Design Within Concept of software Engineering, Design Process- Software Quality guidelines and attributes, The Evolution of Software design, Design Concepts, The Design Model.
**Software Testing Strategies**: Strategic Approach to Software Testing- Verification and validation, Organizing Software testing, Software testing Strategies, Criteria for completion of testing, Strategic Issues, Test Strategy for conventional and Object-Oriented Software - Unit and Integrating testing, Validation
Testing- Validation test Criteria, Configuration review, Alpha and Beta Testing, System Testing – Recovery, Security, Stress, Performance and deployment testing.

**TB1:8.1-8.4,17.1-17.7** **RBT: 6** **WK: 5,8**

## Module 4

**Principles that guide practice**: Software Engineering Knowledge, Core principles, Principles that guide each framework activity.

**Introduction to Project Management**: Introduction, Project and Importance of Project Management, Contract Management, Activities Covered by Software Project Management, Plans, Methods and Methodologies, some ways of categorizing Software Projects, Stakeholders, Setting Objectives, Business Case, Project Success and Failure, Management and Management Control, Project Management life cycle, Traditional versus Modern Project Management Practices.

Overview of Project planning: Steps 0-9 project planning.

**TB1:4.1-4.3 , TB2:1.1-1.17,3.1-3.11** **RBT: 4** **WK: 5,7**

## Module 5

**Project Evaluation and Program Management**: Introduction, Business Case, Project Portfolio Management, Evaluation of Individual projects, Cost–benefit Evaluation Techniques, Risk Evaluation, Program Management, Managing Allocation of resources within Programs, Strategic Program Management, Aids to program Management, Reservations of Program Management and Benefits of Management.

**Software Project Estimation**: Observations on Estimation, Project Planning Process, Software Scope and feasibility, Resources, Software Project Estimation, Decomposition Techniques, Empirical Estimation Models, Estimation for Object Oriented Projects, Specialized Estimation Techniques, The Make or Buy Decision.

**TB1:26.1-26.10 , TB2:2.1-2.9,2.11-2.13** **RBT: 2** **WK: 5**

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 1. | Roger S. Pressman | Software Engineering-A Practitioners approach | McGraw-Hill, 7th Edition | 2010 |
| 2. | Bob Hughes, Mike Cotterell, Rajib Mall | Software Project Management | McGraw Hill Education, 6th Edition | 2018 |

## References

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 1. | Ian Sommerville | Software Engineering | Pearson, 9th Edition | 2009 |
| 2. | Pankaj Jalote | An Integrated Approach to Software Engineering | Springer, 3rd Edition | 2005 |
| 3. | Hans van Vliet | Software Engineering: Principles and Practice | Wiley India, 3rd Edition | 2010 |

## Course Outcomes

**At the end of the course, the student will be able to**:

1. **Demonstrate** the basic knowledge of software development life cycle processes.
2. **Apply** Requirement Engineering activity and construct the SRS for varied types of software's based on their technical paradigms.
3. **Analyze** and **apply** the software design techniques and build test cases to develop structured software solutions.
4. **Illustrate** the role of project planning and project management in software development.
5. **Interpret** the importance of software project management activities involving time, cost, and resource scheduling.

## Web Links

1. https://onlinecourses.nptel.ac.in/noc20_cs68/preview
2. https://onlinecourses.nptel.ac.in/noc24_mg01/preview

## Activity-Based Learning

1.Demonstration of Agile tool:

The students are expected to learn any of the popular agile tool. (10 marks)

2. Field Survey (In Team):

i. The students' team may of the size of 2 or 4. Students are expected to visit their library and understand the Library Automation Software. OR they have to understand the working of ERP or any inventory management.

ii. Prepare a document/report which includes all the phases of SDLC and to be submitted accordingly (15 marks).

## CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | | | | | | 3 | 1 | 2 | | 3 | |
| CO2 | 3 | 3 | 3 | | 3 | | | 2 | 3 | | | 3 | |
| CO3 | | 3 | 2 | 3 | 3 | | | | | | | 3 | 1 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO4 | | | | | | 1 | | 2 | 3 | 2 | 1 | 2 | 3 |
| CO5 | | | | | | 3 | | 1 | 3 | 3 | 1 | 3 | 2 |
| **AVG** | **3.0** | **2.7** | **2.5** | **3.0** | **3.0** | **2.0** | | **2.0** | **2.5** | **2.3** | **1.0** | **2.8** | **2.0** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | COMPUTER NETWORKS AND SECURITY | BCS502 | 4 | 40+10 | 3:0:2:0 | 3 | 50 | 50 | IPCC | - |

## Course Objectives

**This course will enable the students to:**

1. Demonstration of application layer protocols
2. Discuss transport layer services and understand UDP and TCP protocols
3. Explain routers, IP and Routing Algorithms in network layer
4. To analyze different Cryptographic Algorithms
5. Illustrate concepts of Network Security and Management

## Teaching–Learning Process

**In addition to the traditional chalk and talk method, ICT tools are adopted:**

1. Lecturer method (L) need not to be only traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video/Animation to explain functioning of various concepts.
3. Encourage collaborative (Group Learning) Learning in the class.
4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
5. Adopt Problem Based Learning (PBL), which fosters student's Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.

### Module 1

**Introduction**: Data Communications, Networks, **Network Models**:Protocol Layering, TCP/IP Protocol Suite, OSI Model, **Error Detection and Correction**: Introduction, Block Coding, Cyclic Codes, Check Sum, **Media Access Control**: Random Access, Controlled Access.

**TB1:1.1-1.2, 2.1-2.3, 10.1-10.4,12.1-12.2** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 2

**Network Layer**: Network Layer Services, Packet Switching, Network Layer Performance, IPV4 Addresses,
**Network Layer Protocols**: Internet Protocol, Unicast Routing, Routing algorithms, Internet Structure, Routing Information Protocol (RIP), **Next Generation IP**: IPV6 Addressing, IPV6 Protocol, Transition from IPV4 to IPV6.

**TB1:18.1-18.4, 19.1, 20.2- 20.3.2, 22.1-22.2, 22.4** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 3

**Introduction to Transport Layer**: Introduction, Transport-Layer Protocols, **Transport-Layer Protocols**: Introduction, User Datagram Protocol, **Transmission Control Protocol**: services, features, segments, TCP connections, flow control, Error control, Congestion control.

**TB1:23.1,23.2,24.1,24.2,24.3-24.3.1-24.3.4,24.3.6-24.3.9** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 4

**Application Layer**: Introduction, Client-Server Programming, **Standard Client Server Protocols**:World Wide Web and HTTP, FTP, Electronic Mail, TELNET, Secure Shell (SSH), Domain Name System (DNS)

**TB1:25.1-25.2, 26.1- 26.6 (till 26.6.3)** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 5

**Network Security*: Overview of Network Security, Security Methods ,Symmetric-Key Cryptography, Public-Key Cryptography, Authentication, Digital Signatures, Security of IP and Wireless Networks, Firewalls and Packet Filtering.

**TB2:10** **RBT: 1,2,3** **WK: 1,2,3,4**

## Practical Components

| Sl. No. | Experiment |
|---|---|
| 1 | Develop a program to find the shortest path between vertices using the Bellman-Ford and path vector routing algorithm. |
| 2 | Using TCP/IP sockets, write a client – server program to make the client send the file name and to make the server send back the contents of the requested file if present. |
| 3 | Develop a program on a datagram socket for client/server to display the messages on client side, typed at the server side. |
| 4 | Develop a program for a simple RSA algorithm to encrypt and decrypt the data. |
| 5 | Develop a program for congestion control using a leaky bucket algorithm. |
| 6 | Develop a program to implement the Diffie–Hellman Key Exchange Algorithm. |

| Sl. No. | Experiment |
|---|---|
| 7 | Implement three nodes point – to – point network with duplex links between them. Set the queue size, vary the bandwidth, and find the number of packets dropped. |
| 8 | Implement transmission of ping messages/trace route over a network topology consisting of 6 nodes and find the number of packets dropped due to congestion. |
| 9 | Implement an Ethernet LAN using n nodes and set multiple traffic nodes and plot congestion window for different source / destination |
| 10 | To simulate a Denial-of-Service (DoS) attack in a network using NS2 and analyze its impact on throughput, delay, and packet loss. |

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 1 | Behrouz A. Forouzan | Data Communications and Networking | 5th Edition, Tata McGraw- Hill | 2013 |
| 2 | Nader F Mir | Computer and Communication Networks | 2nd Edition, Pearson Publication | 2014 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Understand** the fundamental needs of communication networks, functionalities of TCP/IP layers and Security Concepts.
2. **Apply** routing **algorithms**, and congestion control methods to ensure reliable data communication.
3. **Implement** different Cryptographic **Algorithms** for different **applications**
4. **Analyse** the various protocols, techniques in TCP/IP network architecture
5. **Explore** and document advanced and emerging topics in networks security and **application** layer services and modern innovations.

## Web Links

1. https://www.digimat.in/nptel/courses/video/106105183/L01.html
2. http://www.digimat.in/nptel/courses/video/106105081/L25.html
3. https://nptel.ac.in/courses/10610
4. https://archive.nptel.ac.in/course.html

## Activity-Based Learning

1. Implementation of various protocols using open source simulation tools.
2. Group assignment to implement Cryptographic Algorithms.

### CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | | | | | | | | | | 2 | 2 |
| CO2 | 3 | 2 | | | | | | | | | | 2 | 1 |
| CO3 | 3 | 2 | | | | | | | | | | 2 | 2 |
| CO4 | 3 | 3 | | | | | | | | | | 2 | 1 |
| CO5 | 2 | 2 | | | 2 | | | 3 | 3 | | 2 | 2 | 1 |
| **AVG** | **2.8** | **2.2** | | | **2.0** | | | **3.0** | **3.0** | | **2.0** | **2.0** | **1.4** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | THEORY OF COMPUTATION | BCS503 | 4 | 50 | 4-0-0-0 | 3 | 50 | 50 | PCC (T) | Theory |

## Course Objectives

**This course will enable the students to:**

1. Introduce core concepts in Automata and Theory of Computation.
2. Identify different Formal Language Classes and their Relationships.
3. Learn concepts of Grammars and Recognizers for different formal languages.
4. Prove or disprove theorems in automata theory using their properties

## Teaching–Learning Process

1. Encourage active learning through problem-solving sessions involving DFA/NFA construction, grammar derivation, and automata minimization techniques.
2. Encourage students to use standard simulation tools such as JFLAP for better visualization and understanding of computational models.
3. Encourage continuous self-learning and practice to improve problem-solving skills related to formal languages and automata theory.
4. Encourage peer learning and group discussions for solving language recognition and automata design problems.
5. Conduct quizzes, assignments, and seminars on computability and complexity concepts to strengthen theoretical understanding.
6. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.

## Modern Tools

1. Jflap
2. AutomataVerse
3. AlgoMaster

### Module 1

**Introduction to Automata Theory:** Central Concepts of Automata Theory, Deterministic Finite Automata (DFA), Non-Deterministic Finite Automata (NFA), Epsilon - NFA, NFA to DFA Conversion, Minimization of DFA

**TB1:1.5,2.2,2.3,2.5,4.4** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 2

**Regular Expressions and Languages:** Regular Expressions, Finite Automata and Regular Expressions, Applications of Regular Expressions, Proving Languages not to be Regular, Closure Properties of Regular Languages

**TB1:3.1,3.2(Except 3.2.1),3.3,4.1,4.2** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 3

**Context-Free Grammars and Languages:** Definition of CFGs, Derivations Using a Grammar, Parse Trees, Ambiguity in Grammars and Languages, Normal Forms for Context-Free Grammars, The Pumping Lemma for Context-Free Languages, and Closure Properties of Context-Free Languages.

**TB1:5.1.1- 5.1.6,5.2(5.2.1,5.2.2),5.4(5.4.1),7.1-7.3** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 4

**Pushdown Automata:** Definition of Pushdown Automata, Languages of a PDA, Equivalence of PDA's and CFG's, Deterministic Pushdown Automata.

**TB1:6.1-6.4** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 5

**Introduction to Turing Machines:** Definition and working model of Turing Machine, TM problems, Programming Techniques for Turing Machines, Extensions to the Basic Turing Machine.

**TB1: 8.1-8.4** **RBT: 1,2,3** **WK: 1,2,3,4**

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 1 | John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman | Introduction to Automata Theory, Languages and Computation | Pearson 3rd Ed. | 2006 |

## References

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | John C. Martin | Introduction to Languages and The Theory of Computation | Tata McGraw-Hil 3rd Ed. | 2013 |
| 2 | K.L.P Mishra, N. Chandrashekaran | Theory of Computer Science | PHI Learning,3rd Edition | 2012 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Understand** the fundamental concepts of Automata Theory and Theory of Computation.
2. **Build** DFA, NFA, and ε-NFA, convert one automata model into another, minimize DFA, and Illustrate the properties of Regular Languages.
3. **Develop** Context-Free Grammars (CFGs) and Pushdown Automata (PDAs) for Formal Languages and Interpret the properties of Context Free Languages (CFLs).
4. **Construct** Turing Machines to solve the computational problems.

## Web Links

1. https://www.youtube.com/playlist?list=PL3-wYxbt4yCgBHUpwXDTLos3JStccGIax
2. https://www.youtube.com/watch?v=T5twacMzBFo

## Activity-Based Learning

1. Open source tools (like JFLAP) to make teaching and learning more interactive https://www.jflap.org/] (10 Marks)
2. Assignments at RBTL- 4 (15 marks)

### CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | | | | | | | | | | | | |
| CO2 | 3 | 3 | 3 | | 3 | | | 2 | 2 | | | 3 | 2 |
| CO3 | 3 | 3 | 3 | | 3 | | | 2 | 2 | | | 3 | 2 |
| CO4 | 3 | 3 | 3 | | 3 | | | 2 | 2 | | | 3 | 2 |
| **AVG** | **3.0** | **3.0** | **3.0** | | **3.0** | | | **2.0** | **2.0** | | | **3.0** | **2.0** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | WEB TECHNOLOGY LAB | BCSL504 | 1 | 12 | 0:0:2:0 | 2 | 50 | 50 | PCCL | - |

## Course Objectives

**This course will enable the students to:**

1. Learn HTML 5 elements and their use.
2. Use of CSS for enhanced user interface presentation.
3. Gain knowledge of JavaScript, AJAX and jQuery for dynamic presentation.
4. Gain knowledge of React JS applications.
5. Design and develop Websites and Web applications.

## Teaching–Learning Process

**In addition to the traditional chalk and talk method, ICT tools are adopted:**

1. Concepts' explanation in laboratory.
2. Program explanation in laboratory.

## Modern Tools

**Modern AI tools used for this course:**

1. VS Code

## Practical Components

| Sl. No. | Experiment |
|---|---|
| 1 | Develop the HTML page named "Myfirstwebpage.html". Add the following tags with relevant content:<br>1. Set the title of the page as "My First Web Page"<br>2. Within the body use the following tags:<br>a) Moving text = "Basic HTML Tags"<br>b) Different heading tags (h1 to h6)<br>c) Paragraph<br>d) Horizontal line<br>e) Line Break<br>f) Block Quote<br>g) Pre tag<br>h) Different Logical Style (<b>, <u>, <sub>, <sup> etc.) |
| 2 | Develop the HTML page named "Table.html" to display your class time table.<br>a) Provide the title as Time Table with table header and table footer, row-span and col-span etc.<br>b) Provide various colour options to the cells (Highlight the lab hours and elective hours with different colours.)<br>c) Provide colour options for rows. |
| 3 | Develop an external style sheet named "style.css" and provide different styles for h2, h3, hr, p, div, span, time, img & a tags. Apply different CSS selectors for tags and demonstrate the significance of each. |
| 4 | Develop HTML page named "registration.html" having variety of HTML input elements with background colors, table for alignment & provide font colors & size using CSS styles. |
| 5 | Develop HTML page named "newpaper.html" having variety of HTML semantic elements with background colors, text-colors & size for figure, table, aside, section, article, header, footer... etc. |
| 6 | Apply HTML, CSS and JavaScript to design a simple calculator to perform the following operations: sum, product, difference, remainder, quotient, power, square-root and square. |
| 7 | Develop JavaScript program (with HTML/CSS) for:<br>a) Converting JSON text to JavaScript Object<br>b) Convert JSON results into a date<br>c) Converting From JSON To CSV and CSV to JSON<br>d) Create hash from string using crypto.createHash() method (Note: "node.js" is handwritten next to this) |
| 8 | Develop React application to:<br>a) Display issue details in a tabular format and allow the user to add new issues through a form. Use React Components to manage list of issues.<br>b) Simulate an Issue Tracker system. The application should display issue details in a table and demonstrate data loading asynchronously using the component lifecycle methods. |
| 9 | Develop jQuery script (with HTML/CSS) for:<br>a) Appends the content at the end of the existing paragraph and list.<br>b) Change the state of the element with CSS style using animate() method<br>c) Change the color of any div that is animated. |
| 10 | Develop a JavaScript program with Ajax (with HTML/CSS) for:<br>a) Use ajax() method (without Jquery) to add the text content from the text file by sending ajax request.<br>b) Use ajax() method (with Jquery) to add the text content from the text file by sending ajax request.<br>c) Illustrate the use of getJSON() method in jQuery<br>d) Illustrate the use of parseJSON() method to display JSON values. |

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 1 | Randy Connolly and Ricardo Hoar | Fundamentals of Web Development | Pearson, 3rd Edition | 2021 |
| 2 | Robert W Sebesta | Programming the World Wide Web | Pearson Education, 8th Edition | 2020 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Design** the experiment for the given problem using HTML, Javascript and CSS.
2. **Develop** the solution for the given real-world problem using jQuery, Ajax and React JS
3. **Analyze** runtime results and produce substantial written documentation.

## Web Links

1. https://www.w3schools.com/html/default.asp
2. https://www.w3schools.com/css/default.asp
3. https://www.w3schools.com/js/js_examples.asp
4. https://www.geeksforgeeks.org/javascript-examples/
5. https://www.w3schools.com/react/default.asp
6. https://www.w3schools.com/jquery/default.asp
7. https://www.geeksforgeeks.org/jquery-tutorial/
8. https://www.w3schools.com/js/js_ajax_intro.asp

### CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 3 | | 3 | 2 | 2 | 2 | | | | 2 | 2 |
| CO2 | 3 | 3 | 3 | | 3 | 2 | 2 | 2 | | | | 2 | 2 |
| CO3 | | 3 | | 3 | | | | | 3 | | | | |
| **AVG** | **3.0** | **3.0** | **3.0** | **3.0** | **3.0** | **2.0** | **2.0** | **2.0** | **3.0** | | | **2.0** | **2.0** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | COMPUTER GRAPHICS | BCS515A | 3 | 40 | 3:0:0:0 | 3 | 50 | 50 | PEC (T) | Theory |

## Course Objectives

**This course will enable the students to:**

1. Understand the basic principles of Graphical Systems and OpenGL Graphics Primitives..
2. Design and implementation of algorithms for 2D graphics Primitives and attributes.
3. Demonstrate Geometric transformations, viewing on both 2D and 3D objects.
4. Infer the representation of lines, surfaces, Color and Illumination models.
5. Demonstrate rendering of the curves and surfaces.

## Teaching–Learning Process

**In addition to the traditional chalk and talk method, ICT tools are adopted:**

1. Encourage collaborative (Group Learning) Learning in the class.
2. Demonstrate every concept by implementing an OpenGL program.

## Modern Tools

**Modern AI tools used for this course:**

1. Kittl - AI Design Platform.
2. 3Dify - AI-based 3D Graphics Tools.

### Module 1

**Graphics Systems and Models:** Applications of Computer Graphics, A Graphics System, Images: Physical and Synthetic, Imaging Systems, The Synthetic-Camera Model, The Programmer's Interface, Graphics Architectures, Programmable Pipelines, Performance Characteristics.

**Input and Interaction:** Interaction, Input devices, Clients and Servers, Display Lists, Display Lists and Modeling, Programming Event Driven Input, Menus.

**TB1:1, 3.1-3.7** **RBT: 1, 2, 3** **WK: 1, 3**

### Module 2

**Geometric Objects and Transformations:** Frames in OpenGL, Modeling a Colored Cube, Affine Transformations, Rotation, Translation and Scaling, Transformation in Homogeneous Coordinates, Concatenation of Transformations.

**TB1: 4.4 - 4.9** **RBT: 1, 2, 3** **WK: 1, 2, 3, 4**

### Module 3

**Viewing:** Classical and Computer Viewing, Viewing with a Computer.
**Lighting and Shading:** Light and Matter, Light Sources, The Phong Lighting Model, Polygonal Shading.

**TB1: 5 – 5.1-5.2, 6.1-6.3 & 6.5** **RBT: 2, 3** **WK: 1, 2, 3, 4**

### Module 4

**From Vertices to Fragments:** Basic Implementation Strategies, Four major tasks, Clipping, Line-segment clipping, Cohen-Sutherland Clipping, Liang-Barsky Clipping.

**Implementation Algorithms for Graphics Primitives and Attributes:** Line-Drawing Algorithms, DDA lgorithm, Bresenham's Line Algorithm, Parallel Line Algorithms, Setting Frame-Buffer Values, Circle-Generating Algorithms, Midpoint Circle Algorithm.

**TB1:7.1 -7.4 , TB2:5-1-5-4** **RBT: 2, 3, 4** **WK: 1, 2, 3, 4**

### Module 5

**Programmable Shaders:** Programmable pipelines, Shading Languages, Extending OpenGL, The OpenGL Shading Language, Linking Shaders with OpenGL Programs.

**Advanced Rendering:** Going beyond pipeline rendering, Ray Tracing, Building a Simple Ray Tracer.

**TB1:9.1-9.6,13.1-13.3** **RBT: 2, 3** **WK: 1, 2, 3, 4, 6**

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 1 | Edward Angel | Interactive Computer Graphics A Top-Down Approach with OpenGL | Pearson Education & 5th Edition | 2008 |
| 2 | Donald Hearn & Pauline Baker | Computer Graphics with OpenGL Version | Pearson Education & 4th Edition | 2011 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Understand** the fundamentals of computer graphics systems.
2. **Apply** geometrical transformations along with viewing, lighting, and shading techniques on graphical objects.
3. **Develop** event-driven graphical **applications** incorporating various **algorithms** and object rendering.
4. **Demonstrate** the **application** and explain its functionality.

### CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | | | | | | | | | | | |
| CO2 | 3 | 2 | 2 | | | | | | | | | 2 | |
| CO3 | 3 | 2 | 2 | | | | | | | | | 2 | |
| CO4 | 3 | 3 | 3 | 2 | 3 | 2 | 2 | 3 | 3 | | | 2 | 2 |
| **AVG** | **3.0** | **2.3** | **2.3** | **2.0** | **3.0** | **2.0** | **2.0** | **3.0** | **3.0** | | | **2.0** | **2.0** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | ARTIFICIAL INTELLEGENCE | BCS515B | 3 | 40 | 3:0:0:0 | 3 | 50 | 50 | PEC (T) | Theory |

## Course Objectives

**This course will enable the students to:**

1. Learn the basic principles and theories underlying artificial intelligence, including machine learning, neural networks, natural language processing, and robotics
2. Apply AI techniques to solve real-world problems, including search algorithms, optimization, and decision-making processes.
3. Understand the ethical, legal, and societal implications of AI, including topics such as bias, fairness, accountability, and the impact of AI on the workforce and privacy

## Teaching–Learning Process

**In addition to the traditional chalk and talk method, ICT tools are adopted:**

1. Use of Video/Animation to explain functioning of various concepts.
2. Encourage collaborative (Group Learning) Learning in the class.
3. Discuss application of every concept to solve the real-world problems.
4. Demonstrate ways to solve the same problem and encourage the students to come up with their own creative solutions.

## Modern Tools

1. Perplexity AI – Research-oriented explanations with references
2. Canva AI / PowerPoint Copilot – AI-assisted presentations and diagrams
3. Kahoot AI / Quizizz AI – AI-generated quizzes for formative assessment
4. Grammarly AI – Technical writing improvement for assignments

### Module 1

**Introduction**: What Is AI? , The State of The Art.
**Intelligent Agents**: Agents and environment, Concept of Rationality, The nature of environment, The structure of agents

**TB1:1.1, 1.4, 2.1, 2.2, 2.3, 2.4** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 2

**Problem-solving**: Problem-solving agents, Example problems, Searching for Solutions Uninformed Search Strategies

**TB1: 3.1, 3.2, 3.3, 3.4** **RBT: 1,2,3,4** **WK: 1,2,3,4**

### Module 3

**Problem-solving**: Informed Search Strategies, Heuristic functions
**Logical Agents**: Knowledge–based agents, The Wumpus world, Logic, Propositional logic, Reasoning patterns in Propositional Logic

**TB1:3.5, 7.6 ,7.1, 7.2, 7.3, 7.4** **RBT: 1,2,3,4** **WK: 1,2,3,4,**

### Module 4

**First Order Logic**: Representation Revisited, Syntax and Semantics of First Order logic, Using First Order logic, Knowledge Engineering in First-Order Logic.
**Inference in First Order Logic**: Propositional Versus First Order Inference, Unification,
Forward Chaining

**TB1:8.1, 8.2, 8.3, 8.4,9.1, 9.2, 9.3** **RBT: 1,2,3** **WK: 1,2,3,4**

### Module 5

**Inference in First Order Logic**: Backward Chaining, Resolution
**Classical Planning**: Definition of Classical Planning, Algorithms for Planning as State-Space Search, Planning Graphs.

**TB1:9-9.4, 9.5,10.1,10.2,10.3** **RBT: 1,2,3** **WK: 1,2,3,4**

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 01 | Stuart J. Russell and Peter Norvig, | Artificial Intelligence | Pearson, 3rd Edition | 2015 |

## References

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 01 | Elaine Rich, Kevin Knight | Artificial Intelligence, | Tata McGraw Hill,3rd Edition | 2013 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Understand** the basic concepts of AI - searching, reasoning, knowledge
2. **Apply** knowledge of searching, reasoning and first order for solving problems.
3. **Analyze** various Searching and Inferencing Techniques for a given scenario..
4. **Design** a Chatbot using any Open AI tool.

## Web Links

1.https://www.kdnuggets.com/2019/11/10-free-must-read-books-ai.html
2. https://www.udacity.com/course/knowledge-based-ai-cognitive-systems--ud409
3. https://nptel.ac.in/courses/106/105/106105077/

### CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | | | | | | | | | | | 1 | 2 |
| CO2 | 3 | 3 | 2 | | | | | | | | | 1 | 2 |
| CO3 | 3 | 3 | 2 | | 2 | | | | 3 | 3 | | 1 | 2 |
| CO4 | 2 | 2 | 3 | | 2 | | | | 3 | 3 | | 1 | 2 |
| **AVG** | **2.5** | **2.7** | **2.3** | | **2.0** | | | | **3.0** | **3.0** | | **1.0** | **2.0** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | UNIX SYSTEM PROGRAMMING | BCS515C | 3 | 40 | 3:0:0:0 | 3 | 50 | 50 | PEC (T) | Theory |

## Course Objectives

**This course will enable the students to:**

1. To help the students to understand effective use of Unix concepts, commands and terminology. Identify, access, and evaluate UNIX file system.
2. Explain the fundamental design of the Unix operating system.
3. Familiarize with the systems calls provided in the Unix environment.
4. Design and build an application/service over the Unix operating system.

## Teaching–Learning Process

**In addition to the traditional chalk and talk method, ICT tools are adopted:**

1. Lecturer method (L) need not to be only a traditional lecture method, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video/Animation to explain functioning of various concepts.
3. Encourage collaborative (Group Learning) Learning in the class.
4. Ask at least three HOT (Higher order Thinking) questions in the class, which promotes critical thinking.
5. Adopt Problem Based Learning (PBL), which fosters students' Analytical skills, develop design thinking skills such as the ability to design, evaluate, generalize, and analyze information rather than simply recall it.

## Modern Tools

**Modern AI tools used for this course:**

1. AI Coding Assistants: GitHub Copilot - AI-powered code completion inside editors like VS Code, Vim, Neovim.
2. AI-Powered Development Environments: Cursor - AI-native code editor.
3. AI for Command Line & Terminal: Warp, Fig.
4. AI Debugging & Code Analysis: DeepCode, SonarQube.
5. AI for Documentation & Learning: ExplainShell.

### Module 1

**Introduction**: Unix Components/Architecture. Features of Unix. The UNIX Environment and UNIX Structure, Posix and Single Unix specification. General features of Unix commands/ command structure. Command arguments and options. Basic Unix commands such as echo, printf, ls, who, date, passwd, cal, Combining commands. Meaning of Internal and external commands. The type command: knowing the type of a command and locating it. The root login. Becoming the super user: su command.

**Unix files**: Naming files. Basic file types/categories. Organization of files. Hidden files. Standard directories. Parent-child relationship. The home directory and the HOME variable. Reaching required files- the PATH variable, manipulating the PATH, Relative and absolute pathnames. Directory commands – pwd, cd, mkdir, rmdir commands. The dot (.) and double dots (..) notations to represent present and parent directories and their usage in relative path names. File related commands – cat, mv, rm, cp, wc and od commands.

**TB1:1, 2, 3, 4, 5** **RBT: 2, 3** **WK: 3**

### Module 2

**File attributes and permissions**: The ls command with options. Changing file permissions: the relative and absolute permissions changing methods. Recursively changing file permissions. Directory permissions.
**The shells interpretive cycle**: Wild cards. Removing the special meanings of wild cards. Three standard files and redirection.
**Connecting commands**: Pipe. Basic and Extended regular expressions. The grep, egrep. Typical examples involving different regular expressions.

**Shell programming**: Ordinary and environment variables. The profile Read and read-only commands. Command line arguments. exit and exit status of a command. Logical operators for conditional execution. The test command and its shortcut. The if, while, for and case control statements. The set and shift commands and handling positional parameters. The here (<<) document and trap command. Simple shell program examples.

**TB1:6, 8, 13, 14** **RBT: 3, 4** **WK: 4**

## Module 3

**Unix Standardization and Implementations**: Introduction, Unix Standardization, UNIX System Implementation.
**File I/O**: Introduction, File Description, open, create, read, write, close, fcntl functions.
**Files and Dictionaries**: mkdir and rmdir functions, reading dictionaries, chdir, fchdir and getcwd functions. Device Special files.
**The Environment of a UNIX Process**: Introduction, main function, Process Termination, Command-Line Arguments, Environment List, Memory Layout of a C Program, Shared Libraries, Memory Allocation, Environment Variables, setjmp and longjmp Functions, getrlimit, setrlimit Functions.

**TB2:2, 3, 4, 7** **RBT: 2, 3, 4** **WK: 4**

## Module 4

**Process Control**: Introduction, Process Identifiers, fork, vfork, exit, wait, waitpid, wait3, wait4 Functions, Race Conditions, exec Functions.
**Overview of IPC Methods**: Pipes, popen, pclose Functions, Coprocesses, FIFOs, System V IPC, Message Queues, Semaphores.
**Shared Memory**: Client-Server Properties, Passing File Descriptors, An Open Server-Version 1.

**TB2:8, 15, 17** **RBT: 3, 4, 5** **WK: 4**

## Module 5

**Signals and Daemon Processes**: Introduction, Signal Concepts, Signal Functions, SIGCLD Semantics, Kill and Raise functions, Alarm and Pause Functions, Signal Sets, sigprocmask Function, sigpending function, sigaction function, sigsetjmp and siglongjmp functions, sigsuspend function, abort function, system function, sleep, nanosleep and clock_nanosleep functions, sigqueue functions, job-control signals, signal names and numbers.
**Daemon Processes**: Introduction, Daemon Characteristics, Coding Rules, Error Logging, Client-Server Model.

**TB2:10, 13** **RBT: 4, 5** **WK: 4,5**

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 1 | Sumitabha Das | Unix Concepts and Applications | Tata McGraw Hill | 2017 |
| 2 | W. Richard Stevens | Advanced Programming in the UNIX Environment | Pearson Education | 2005 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Demonstrate** the basics of Unix concepts and commands.
2. **Demonstrate** Unix file system.
3. **Apply** commands to reflect changes in file system.
4. **Demonstrate** IPC and process management.
5. **Develop** an **application**/service over a Unix system.

## Web Links

1. https://www.youtube.com/watch?v=ffYUfAqEamY
2. https://www.youtube.com/watch?v=Q05NZiYFcD0
3. https://www.youtube.com/watch?v=8GdT53KDIyY
4. https://www.youtube.com/watch?app=desktop&v=3Pga3y7rCgo

## Activity-Based Learning

1. Programming assignment -1 (Shell level) - 10 marks
2. Programming assignment -2 (API level) - 15 marks

## CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | | | 2 | | | | | | | 1 | |
| CO2 | 3 | 2 | | | 2 | | | | | | | 1 | |
| CO3 | 3 | 3 | 2 | | 3 | | | | | | | 1 | 1 |
| CO4 | 2 | 3 | 2 | 2 | 3 | | | | 1 | | | 2 | 2 |
| CO5 | 3 | 3 | 3 | 2 | 3 | 1 | 1 | | 2 | 2 | 2 | 3 | 3 |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **AVG** | **2.8** | **2.6** | **2.3** | **2.0** | **2.6** | **1.0** | **1.0** | | **1.5** | **2.0** | **2.0** | **1.6** | **2.0** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | NATURAL LANGUAGE PROCESSING | BCS515D | 3 | 40 | 3:0:0:0 | 3 | 50 | 50 | PEC (T) | Theory |

## Course Objectives

**This course will enable the students to:**

1. Learn the importance of natural language modelling
2. Understand the applications of natural language processing
3. Study spelling, error detection and correction methods and parsing techniques in NLP
4. Illustrate the information retrieval models in natural language processing.

## Teaching–Learning Process

**In addition to the traditional chalk and talk method, ICT tools are adopted:**

1. Power point presentation
2. Promote collaborative learning (Group Learning) in the class through practical based assignments
3. Demonstrate various ways to solve the same problem and encourage students to devise their own creative solutions

## Modern Tools

**Modern AI tools used for this course:**

Students will perform practical assignments using Python packages with the help of Integrated Development Environments such as PyCharm, Visual Studio Code, or Jupyter Notebook Encourages hands-on learning

Helps students use real programming tools Develops practical coding skills with Python

### Module 1

**Introduction**: What is Natural Language Processing? Origins of NLP, Language and Knowledge, The Challenges of NLP, Language and Grammar, Processing Indian Languages, NLP Applications.
**Language Modeling**: Statistical Language Model - N-gram model (unigram, bigram),
Paninion Framework, Karaka theory

**TB1:1,2** **RBT: 1,2,3** **WK: 1,2,3,4,5,8**

### Module 2

**Word Level Analysis**: Regular Expressions, Finite-State Automata, Morphological Parsing, Spelling Error Detection and Correction, Words and Word Classes, Part-of Speech Tagging.
**Syntactic Analysis**: Context-Free Grammar, Constituency, Top-down and Bottom-up Parsing, CYK Parsing.

**TB1:3,4** **RBT: 1,2,3** **WK: 1,2,3,4,5,8**

### Module 3

Naive Bayes, Text Classification and Sentiment: Naive Bayes Classifiers, Training the Naive Bayes Classifier, Worked Example, Optimizing for Sentiment Analysis, Naive Bayes for Other Text Classification Tasks, Naive Bayes as a Language Model

**TB2:4** **RBT: 1,2,3** **WK: 1,2,3,4,5,8**

### Module 4

Information Retrieval**: Design Features of Information Retrieval Systems, Information Retrieval Models - Classical, Non-classical, Alternative Models of Information Retrieval - Custer model, Fuzzy model, LSTM model, Major Issues in Information Retrieval.
**Lexical Resources**: WordNet, FrameNet, Stemmers, Parts-of-Speech Tagger, Research Corpora

**TB1:9,12** **RBT: 1,2,3** **WK: 1,2,3,4,5,8**

### Module 5

**Machine Translation**: Language Divergences and Typology, Machine Translation using Encoder-Decoder, Details of the Encoder-Decoder Model, Translating in Low-Resource Situations, MT Evaluation, Bias and Ethical Issues

**TB2:13** **RBT: 1,2,3** **WK: 1,2,3,4,5,8**

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 1 | Tanveer Siddiqui, U.S. Tiwary | Natural Language Processing and Information Retrieval | Oxford University Press | - |
| 2 | Daniel Jurafsky, James H. Martin | Speech and Language Processing, An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition | Pearson Education | 2023 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Apply** the fundamental concept of NLP, grammar-based language model and statistical-based language model.

2. **Model** morphological analysis using Finite State Transducers and parsing using context-free grammar and different parsing approaches.

3. **Develop** the Naïve Bayes classifier and sentiment analysis for Natural language problems and text classifications.

4. **Apply** the concepts of information retrieval, lexical semantics, lexical dictionaries such as WordNet, lexical computational semantics, distributional word similarity.

5. **Identify** the Machine Translation **applications** of NLP using Encode and Decoder.

## Web Links

1. https://www.youtube.com/watch?v=M7SWr5xObkA
2. https://youtu.be/02QWRAhGc7g
3. https://www.youtube.com/watch?v=CMrHM8a3hqw
4. https://onlinecourses.nptel.ac.in/noc23_cs45/preview
5. https://archive.nptel.ac.in/courses/106/106/106106211/

## Activity-Based Learning

Text Classification Game (15 Marks)

1. Objective: Learn supervised learning and text classification.

2. Activity: Provide students with a set of documents (e.g., movie reviews) labeled as positive or negative. Divide them into groups and have them create a simple classification model using keywords or phrases. They can then test their model on new reviews.

Grammar Check and Correction (10 Marks)

1.Objective: Learn about language structure and NLP tools.

2. Activity: Provide sentences with grammatical errors. Students can use grammar checking tools (like Grammarly or Language Tool) to identify errors and suggest corrections, discussing why each suggestion is made.

### CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | | | | | | | | | | 2 | 3 |
| CO2 | 3 | 3 | 3 | | | | | | | | | 3 | |
| CO3 | | 3 | 3 | | 3 | | | | | | | 3 | 3 |
| CO4 | | 3 | 3 | 3 | 2 | | | | | | | 2 | 3 |
| CO5 | 2 | 2 | 2 | | 2 | 2 | 2 | 2 | | | | 2 | 2 |
| **AVG** | **2.7** | **2.8** | **2.8** | **3.0** | **2.3** | **2.0** | **2.0** | **2.0** | | | | **2.4** | **2.8** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | RESEARCH METHODOLOGY & IPR | BRMK507 | 3 | 40 | 3:0:0:0 | 3 | 50 | 50 | AEC (T) | Theory |

## Course Objectives

**This course will enable the students to:**

1. Understand the knowledge on basics of research and its types.
2. Learn the concept of Literature Review, Technical Reading, Attributions and Citations.
3. Learn Ethics in Engineering Research.
4. Discuss the concepts of Intellectual Property Rights in engineering

## Teaching–Learning Process

**In addition to the traditional chalk and talk method, ICT tools are adopted:**

1. Lecturer methods (L) need not be only the traditional lecture methods, but alternative effective teaching methods could be adopted to attain the outcomes.
2. Use of Video to explain various concepts on IPR.
3. Encourage collaborative (Group Learning) Learning in the class
4. Show the different ways to analyze the research problem and encourage the students to come up with their own creative ways to solve them.

## Modern Tools

**Modern AI tools used for this course:**

1. Research Rabbit
2. Google Patents
3. Scite

### Module 1

**Introduction**: Meaning of Research, Objectives of Engineering Research, and Motivation in Engineering Research, Types of Engineering Research, Finding and Solving a Worthwhile Problem. Ethics in Engineering Research, Ethics in Engineering Research Practice, Types of Research Misconduct, Ethical Issues Related to Authorship.

**TB1:1** **RBT: 2** **WK: 1,8**

### Module 2

Literature Review and Technical Reading, New and Existing Knowledge, Analysis and Synthesis of Prior Art Bibliographic Databases, Web of Science, Google and Google Scholar, Effective Search: The Way Forward Introduction to Technical Reading Conceptualizing Research, Critical and Creative Reading, Taking Notes While Reading, Reading Mathematics and Algorithms, Reading a Datasheet.

**Attributions and Citations**: Giving Credit Wherever Due, Citations: Functions and Attributes, Impact of Title and Keywords on Citations, Knowledge Flow through Citation, Citing Datasets, Styles for Citations, Acknowledgments and Attributions, What Should Be Acknowledged, Acknowledgments in, Books Dissertations, Dedication or Acknowledgments.

**TB1:2, 3** **RBT: 1,2** **WK: 1,8,9**

### Module 3

**Introduction To Intellectual Property**: Role of IP in the Economic and Cultural Development of the Society, IP Governance, IP as a Global Indicator of Innovation, Origin of IP History of IP in India. Major Amendments in IP Laws and Acts in India.

Patents: Conditions for Obtaining a Patent Protection, To Patent or Not to Patent an Invention. Rights Associated with Patents. Enforcement of Patent Rights. Inventions Eligible for Patenting. Non-Patentable Matters. Patent Infringements. Avoid Public Disclosure of an Invention before Patenting. Process of Patenting.

**Process of Patenting**: Prior Art Search. Choice of Application to be Filed. Patent Application Forms. Jurisdiction of Filing Patent Application. Publication. Pre-grant Opposition. Examination. Grant of a Patent. Validity of Patent Protection. Post-grant Opposition. Commercialization of a Patent. Need for a Patent Attorney/Agent. Can a Worldwide Patent be Obtained? Do I Need First to File a Patent in India? Patent Related Forms. Fee Structure. Types of Patent Applications. Commonly Used Terms in Patenting. National Bodies Dealing with Patent Affairs. Utility Models.

**TB1:4, 5, 6** **RBT: 2** **WK: 1,8,9**

## Module 4

**Copyrights and Related Rights**: Classes of Copyrights. Criteria for Copyright. Ownership of Copyright. Copyrights of the Author. Copyright Infringements. Copyright Infringement is a Criminal Offence. Copyright Infringement is a Cognizable Offence. Fair Use Doctrine. Copyrights and Internet. Non-copyright Work. Copyright Registration. Judicial Powers of the Registrar of Copyrights. Fee Structure. Copyright Symbol. Validity of Copyright. Copyright Profile of India. Copyright and the word 'Publish'. Transfer of Copyrights to a Publisher. Copyrights and the Word 'Adaptation'. Copyrights and the Word 'Indian Work'. Joint Authorship. Copyright Society. Copyright Board. Copyright Enforcement Advisory Council (CEAC). International Copyright Agreements, Conventions and Treaties. Interesting Copyrights Cases.

Trademarks: Eligibility Criteria. Who Can Apply for a Trademark. Acts and Laws. Designation of Trademark Symbols. Classification of Trademarks. Registration of a Trademark is Not Compulsory. Validity of Trademark. Types of Trademark Registered in India. Trademark Registry. Process for Trademarks Registration. Prior Art Search. Famous Case Law: Coca-Cola Company vs. Bisleri International Pvt. Ltd.

**TB1:7, 8** **RBT: 1,2,3** **WK: 1,8,9**

## Module 5

**Industrial Designs**: Eligibility Criteria. Acts and Laws to Govern Industrial Designs. Design Rights. Enforcement of Design Rights. Non-Protectable Industrial Designs India. Protection Term. Procedure for Registration of Industrial Designs. Prior Art Search. Application for Registration. Duration of the Registration of a Design. Importance of Design Registration. Cancellation of the Registered Design. Application Forms. Classification of Industrial Designs. Designs Registration Trend in India. International Treaties. Famous Case Law: Apple Inc. vs. Samsung Electronics Co.

Geographical Indications: Acts, Laws and Rules Pertaining to GI. Ownership of GI. Rights Granted to the Holders. Registered GI in India. Identification of Registered GI. Classes of GI. Non-Registerable GI. Protection of GI. Collective or Certification Marks. Enforcement of GI Rights. Procedure for GI Registration Documents Required for GI Registration. GI Ecosystem in India.

**Case Studies on Patents**:Case study of Curcuma (Turmeric) Patent, Case study of Neem Patent, Case study of Basmati patent. IP Organizations In India. Schemes and Programs.

**TB1:9, 10, 11** **RBT: 2,3** **WK: 1,8,9**

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 1 | Dr. Santosh M Nejakar, Dr. Harish Bendigeri | "Research Methodology and Intellectual Property Rights" | Nejakar Publications 2024 Haveri | 2024 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Understand** the concepts, objectives, types, and ethical practices of engineering research and identify research problems.
2. **Apply** reading techniques for effective scholarly information gathering.
3. **Analyze** prior art and research literature to identify gaps and develop research insight.
4. **Describe** the scope, protection, and significance of different forms of Intellectual Property Rights.
5. **Assess** patentability and legal aspects related to engineering research and innovation.

## Web Links

1. https://youtu.be/SSvZ5UqhkxY?si=H17X9h22nnkJhRQV
2. https://youtu.be/4lnFn5_qeow?si=u4EcO4EJ7WyIYLPH
3. https://youtu.be/abkm0GmIbJQ?si=aGEyyDAEdtF3ma5z
4. https://youtu.be/UhWcAfgtcaI?si=iwmG17dUsoaeWoLC
5. https://youtu.be/wwLowABbEdQ?si=RxomQVthqX-Gp486

## Activity-Based Learning

1. Paper writing using latex Tool
2. Seminar on various case studies & Quizzes

### CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | | | | | 2 | | | | 2 | 2 | |
| CO2 | 2 | 3 | | 2 | 2 | | 2 | | | 2 | 2 | | 3 |
| CO3 | 2 | 3 | | 2 | 2 | | | | | 2 | 2 | | 3 |
| CO4 | 2 | 2 | | | | 2 | 2 | | 2 | 2 | 2 | | 2 |
| CO5 | 2 | 3 | | 2 | | 2 | 2 | | 2 | 2 | 2 | 3 | |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **AVG** | **2.2** | **2.6** | | **2.0** | **2.0** | **2.0** | **2.0** | | **2.0** | **2.0** | **2.0** | **2.5** | **2.7** |

| Sem | Title | Code | Credits | Pedagogy | L-T-P-S | Exam Hrs | CIE | SEE | Course Type | Exam Type |
|---|---|---|---|---|---|---|---|---|---|---|
| 5 | ENVIRONMENTAL STUDIES AND WASTE MANAGEMENT | BESK508 | 1 | 12 | 1:0:0:0 | 1 | 50 | 50 | MC | MCQ |

## Course Objectives

**This course will enable the students to:**

1. Identify the major challenges of environmental issues
2. Develop skills, critical thinking and demonstrate socio-economic skills for Environmental protection
3. Analyze the impact of issues w. r. t. waste management

## Teaching–Learning Process

**In addition to the traditional chalk and talk method, ICT tools are adopted:**

1. Use of Video/Animation to explain functioning of various concepts.
2. Encourage collaborative (Group Learning) Learning in the class.
3. Adopt Case study Based Learning (CBL), which fosters students' analytical skills, develop thinking skills such as the ability to evaluate, generalize, and analyze information rather than simply recall it
4. Discuss how every concept can be applied to the real world - and when that's possible, it helps improve the students' understanding.

### Module 1

**Ecosystem:** Structure of Ecosystem, Types: Forest, Desert, Wetlands, Riverine, Oceanic ecosystems.
**Sustainability:** 17SDG targets and possible actions.
**Self-Study Component (SSC):** Components of the environment.

**TB1:3** **RBT: 1** **WK: 5,7**

### Module 2

**Natural Resources:** Water resources – Availability & Quality aspects, Water borne diseases & water induced diseases, Fluoride problem in drinking water.
**Energy:** Different types of energy, Conventional sources & Non -Conventional sources of Energy, Solar energy, Wind Energy, Hydrogen as an alternative energy
**Self-Study Component (SSC):** Alternative Energy sources

**TB1:2** **RBT: 1** **WK: 5,7**

### Module 3

**Environmental Pollution:** Water Pollution, Noise pollution, Air pollution (Sources, Impacts, Preventive measures and Public Health Aspects.
**Self-Study Component (SSC):** Case studies of air pollution episodes

**TB1:5** **RBT: 2** **WK: 5,7**

### Module 4

**Waste management:** Solid Waste Management , types and sources, functional elements of SWM, Biomedical Waste Management - Sources, Characteristics
**Environmental Legislation:** Solid Waste Management Rules,2016, Biomedical Waste Management Rules, 2016.
**Self-Study Component (SSC):** Case studies on waste management options.

**TB1:6** **RBT: 1** **WK: 5,7**

### Module 5

**E - Waste Management:** E- waste; composition and generation. Global context in e- waste; E-waste pollutants, E waste hazardous properties, Effects of pollutant (E- waste) on human health and surrounding environment, domestic e-waste disposal, Basic principles of E waste management, Component of E waste management. E-waste (Management and Handling) Rules, 2011; and E-Waste (Management) Rules, 2022 - Salient Features and its implications.
**Self-Study Component (SSC):** E-Waste (Management) Amendment Rules, 2023, 2024

**TB1:6 , TB2:2** **RBT: 1** **WK: 5,7**

## Textbooks

| Sl.No | Author | Title | Publisher&Edition | Year |
|---|---|---|---|---|
| 01 | S M Prakash | "Environmental Studies" | Elite Publishing House, Mangalore | 2018 |
| 02 | Hester R.E., and Harrison R.M | Electronic Waste Management. Science, | RSC Publication | 2009 |

## Course Outcomes

**At the end of the course, the student will be able to:**

1. **Comprehend** the principles of ecology and environmental issues pertaining to air, land, and water on a global scale.
2. **Acquire** observation skills for solving problems related to the environment.
3. **Conduct** survey to describe the realities of waste management system.

## Web Links

1. https://sdgs.un.org/goals
2. https://kspcb.karnataka.gov.in/waste-management/biomedical-waste
3. E Waste (Management) Rules, 2022: https://kspcb.karnataka.gov.in/sites/default/files/inline files/E%20Waste%20%28Manage2

## Activity-Based Learning

1. Analysis report of case study specified in the Textbooks and reference books (one per student). (10 marks)
2. Field Survey (In Team): The students' team of the size of 2 to 4 are expected to visit the organization or Industry understand the waste management, utilization of energy, pollution concerns, e-waste handling and other related suggested best practices specified in the syllabus and then submit a detailed visit report to the concerned staff. (15 marks).

### CO–PO–PSO Mapping

| CO | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PSO1 | PSO2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | | | | 2 | 2 | | | | | | |
| CO2 | 1 | 2 | | | | 2 | 2 | | | | | 2 | |
| CO3 | 1 | 2 | | | | 2 | 2 | | | | | 2 | 2 |
| **AVG** | **1.3** | **1.7** | | | | **2.0** | **2.0** | | | | | **2.0** | **2.0** |